

Selection Sort
 $p_1 = a_0 > a_5$ 3 46 24 52 20 13
 $p_2 = a_1 > a_5$ 0 13 24 52 20 46
 $p_3 = a_2 > a_5$ 0 13 20 52 24 46
 $p_4 = a_3 > a_5$ 0 13 20 24 52 46
 $p_5 = a_4 > a_5$ 0 13 20 24 46 52

13 46 24 52 20 9 min=2
Step 1: Swap(arr[i], arr[j])
0 1 2 3 4 5
9 46 24 52 20 13
Step 2: min=5
0 1 2 3 4 5
9 13 24 52 20 46

13 46 24 52 20 9
Step 1: min=0
0 1 2 3 4 5
9 13 46 24 52 20 13
min=j=5
Swap(arr[i], arr[j])
0 1 2 3 4 5
9 46 24 52 20 13

Step 2: i=1
Bubble Sort
13 46 24 52 20 9

Step 1: 13 46 24 52 20 9
24 46 20 52
9 52

Step 2: 13 24 46 20 9 52
20 46 9 46 52

Step 3: 13 24 20 9 46 52
20 24 9 46 52

Step 4: 13 20 9 24 46 52
9 20 24 46 52

Step 5: 13 9 20 24 46 52
9 13 20 24 46 52

i=0 to n-1
Compare(j, j+1) j=0 to n-2

```
public static int[] BubbleSort(int arr[], int n){  
    for(int i = 0; i < n; i++){ // steps or passes  
        for(int j = 0; j < n-i; j++){  
            if(arr[j] > arr[j+1]){  
                int temp = arr[j];  
                arr[j] = arr[j+1];  
                arr[j+1] = temp;  
            }  
        }  
    }  
}
```

